

L1.2 AV Sensors, V2X, HD Maps Quiz ANS

1. Which perception task detects whether an object exists in the environment?

- A) Object Detection
- B) Classification
- C) Tracking
- D) Segmentation

ANS: A) Object Detection

2. Which perception task identifies an object as a traffic sign, traffic light, or pedestrian?

- A) Detection
- B) Classification
- C) Tracking
- D) Localization

ANS: B) Classification

3. Which perception task follows a moving object across time?

- A) Detection
- B) Classification
- C) Object Tracking
- D) Semantic mapping

ANS: C) Object Tracking

4. Which statement correctly describes semantic segmentation?

- A) It assigns one label to an entire image.
- B) It estimates only the distance to each object.
- C) It assigns every pixel to a specific object instance such as car 1 or car 2.
- D) It assigns each pixel to a semantic category such as road, car, or sky.

ANS: D) It assigns each pixel to a semantic category such as road, car, or sky.

5. How does lidar measure distance?

- A) It illuminates a target with pulsed laser light and measures the reflected pulses.
- B) It detects only naturally reflected visible light.
- C) It triangulates the vehicle position using satellites.
- D) It measures wheel rotation with magnetic encoders.

ANS: A) It illuminates a target with pulsed laser light and measures the reflected pulses.

6. Which sensor triangulates the vehicle's position using satellites?

- A) Radar
- B) GPS
- C) Lidar
- D) Ultrasonic sensor

ANS: B) GPS

7. Which sensor uses high-frequency sound waves and their reflections to calculate distance?

- A) Camera
- B) GPS
- C) Ultrasonic sensor
- D) IMU

ANS: C) Ultrasonic sensor

8. What is distinctive about Tesla's current sensor configuration?

- A) It uses lidar but no cameras.
- B) It uses radar as its only perception sensor.
- C) It uses only ultrasonic sensors for close-range perception.
- D) It relies on cameras rather than lidar, radar, or ultrasonic sensors.

ANS: D) It relies on cameras rather than lidar, radar, or ultrasonic sensors.

9. Which of the following is a passive sensor?

- A) Camera
- B) Lidar
- C) Radar
- D) Ultrasonic sensor

ANS: A) Camera

10. What distinguishes an active sensor from a passive sensor?

- A) It can operate only during daylight.
- B) It sends its own signal and senses the reflection.
- C) It always produces a color image.
- D) It determines position only from satellites.

ANS: B) It sends its own signal and senses the reflection.

11. What enables stereo vision in a camera-based system?

- A) One camera and one radar
- B) One rotating lidar

- C) Two cameras
- D) One camera and one ultrasonic sensor

ANS: C) Two cameras

12. What does a camera's field of view (FOV) specify?

- A) The number of images captured per second
- B) The difference between the darkest and brightest pixels
- C) The maximum object-detection distance
- D) The extent of the observable world visible at a given moment

ANS: D) The extent of the observable world visible at a given moment

13. What happens when camera FOV is widened while resolution remains the same?

- A) Image distortion becomes larger.
- B) The camera becomes independent of illumination.
- C) Dynamic range automatically exceeds 200 dB.
- D) The image always contains more pixels.

ANS: A) Image distortion becomes larger.

14. What is camera dynamic range?

- A) The maximum distance at which the camera detects an object
- B) The maximum difference between the darkest and lightest pixel intensities
- C) The angular distance covered by two cameras
- D) The number of frames stored in memory

ANS: B) The maximum difference between the darkest and lightest pixel intensities

15. What does a lidar's point cloud describe?

- A) The color distribution of the road surface
- B) The vehicle's satellite coordinates
- C) The audio spectrum around the vehicle
- D) The 3D geometry of the surrounding environment

ANS: D) The 3D geometry of the surrounding environment

16. Which is an advantage of lidar stated in the presentation?

- A) It provides precise distance measurements and is independent of lighting conditions.
- B) It is always the cheapest AV sensor.

- C) It has higher image classification accuracy than a camera.
- D) It is unaffected by all forms of bad weather.

ANS: A) It provides precise distance measurements and is independent of lighting conditions.

17. Which set contains key lidar parameters?

- A) Color depth, focal length, and aperture
- B) Laser beam count, rotation speed, and range distance
- C) Satellite count, update rate, and map scale
- D) Sound frequency, tire pressure, and wheel speed

ANS: B) Laser beam count, rotation speed, and range distance

18. Which information can mmWave radar provide directly?

- A) Only object color
- B) Only semantic class labels
- C) Position and relative speed
- D) Lane-marking text

ANS: C) Position and relative speed

19. How is long-range automotive radar characterized in the presentation?

- A) Short range with a wide FOV
- B) Up to 200 m with a panoramic FOV
- C) A few meters with a narrow FOV
- D) Up to 250 m with a narrow FOV

ANS: D) Up to 250 m with a narrow FOV

20. What is a common AV application of ultrasonic sensors?

- A) Parking assistance
- B) Long-range highway mapping at 250 m
- C) Traffic-sign classification from color images
- D) Global satellite localization

ANS: A) Parking assistance

21. Why is sensor fusion crucial for robust AV perception?

- A) Every sensor has identical strengths and weaknesses.
- B) Different sensors have complementary strengths and weaknesses.
- C) It eliminates the need for real-time processing.
- D) It guarantees that maps never become outdated.

ANS: B) Different sensors have complementary strengths and weaknesses.

22. Which pair gives examples of Global Navigation Satellite Systems (GNSS)?

- A) Radar and lidar
- B) DSRC and C-V2X
- C) GPS and BeiDou
- D) IMU and RTK

ANS: C) GPS and BeiDou

23. What position accuracy does conventional GPS typically provide?

- A) Submillimeter accuracy
- B) Centimeter-level accuracy under all conditions
- C) Hundreds-of-meters accuracy
- D) A few meters of accuracy

ANS: D) A few meters of accuracy

24. How does RTK GPS improve the rover's position estimate?

- A) By increasing camera resolution
- B) By calculating and transmitting differential correction data via radio
- C) By replacing satellites with ultrasonic sensors
- D) By integrating only gyroscope measurements

ANS: B) By calculating and transmitting differential correction data via radio

25. Which condition can degrade GNSS/GPS performance in an urban environment?

- A) A wide camera FOV
- B) High IMU update frequency
- C) Signal blockage and multipath interference from tall buildings
- D) Using a high-definition map

ANS: C) Signal blockage and multipath interference from tall buildings

26. What do accelerometers in an IMU measure?

- A) Linear acceleration along the x, y, and z axes
- B) The Earth's magnetic field only
- C) Distance from laser pulse reflections
- D) Radio-signal latency to a base station

ANS: A) Linear acceleration along the x, y, and z axes

27. What do gyroscopes in an IMU measure for vehicle motion estimation?

- A) Pixel intensity
- B) Rotational motion around the axes
- C) Satellite ephemeris data
- D) Road-lane topology

ANS: B) Rotational motion around the axes

28. What additional information can a magnetometer provide when included in an IMU?

- A) Object class labels
- B) Long-range depth
- C) Orientation information from the Earth's magnetic field
- D) Traffic-light timing from infrastructure

ANS: C) Orientation information from the Earth's magnetic field

29. What is a major limitation of IMU-based estimation?

- A) It cannot operate in poor lighting.
- B) It produces data only at 10 Hz.
- C) It requires a lidar point cloud to measure acceleration.
- D) It is prone to drift and accumulated errors over time.

ANS: D) It is prone to drift and accumulated errors over time.

30. Which V2X type can support a collision-avoidance system through communication between vehicles?

- A) Vehicle-to-Vehicle (V2V)
- B) Vehicle-to-Pedestrian (V2P)
- C) Vehicle-to-Infrastructure (V2I)
- D) Vehicle-to-Network (V2N)

ANS: A) Vehicle-to-Vehicle (V2V)

31. Which V2X type can send safety alerts to pedestrians and bicyclists?

- A) V2V
- B) V2P
- C) V2I
- D) V2N

ANS: B) V2P

32. Adaptive traffic-light control is an example of which V2X type?

- A) V2V

- B) V2P
- C) V2I
- D) V2N

ANS: C) V2I

33. Real-time traffic routing and cloud services are examples of which V2X type?

- A) V2V
- B) V2P
- C) V2I
- D) V2N

ANS: D) V2N

34. Which technology is based on IEEE 802.11p at 5.9 GHz?

- A) DSRC
- B) RTK GPS
- C) IMU
- D) HD mapping

ANS: A) DSRC

35. For which applications was DSRC presented as particularly suitable?

- A) Offline map acquisition
- B) Latency-sensitive applications
- C) Only entertainment downloads
- D) Satellite navigation correction

ANS: B) Latency-sensitive applications

36. Which is a traditional latency-tolerant C-V2X application?

- A) Immediate emergency braking
- B) A do-not-pass warning
- C) Over-the-Air (OTA) updates
- D) Ultrasonic parking assistance

ANS: C) Over-the-Air (OTA) updates

37. Which two transmission modes are defined for C-V2X?

- A) V2V and V2P
- B) Camera mode and lidar mode
- C) DSRC mode and GPS mode

D) V2X-Cellular and V2X-Direct

ANS: D) V2X-Cellular and V2X-Direct

38. How does V2X-Cellular communicate?

- A) Through a cellular base station such as an eNodeB
- B) Only through direct laser pulses
- C) Without any network infrastructure
- D) Only through traffic-sign cameras

ANS: A) Through a cellular base station such as an eNodeB

39. How does V2X-Direct communicate?

- A) Through a cloud server and base station
- B) Directly from device to device without passing through a base station
- C) Only through a GNSS satellite
- D) By exchanging HD-map files once per year

ANS: B) Directly from device to device without passing through a base station

40. Why can C-V2X improve warning time for a disabled vehicle beyond a blind curve?

- A) It replaces all onboard perception sensors.
- B) It guarantees line-of-sight communication.
- C) Its longer range can deliver the warning message earlier.
- D) It reduces the vehicle's camera resolution.

ANS: C) Its longer range can deliver the warning message earlier.

41. Which consortium defines 5G V2X communications?

- A) 3DPP
- B) IEEE-IMU
- C) HDMA
- D) 5GAA

ANS: D) 5GAA

42. Which HD-map layer contains traffic and incident information?

- A) Dynamic layer
- B) Localization-objects layer
- C) Lane layer
- D) Road layer

ANS: A) Dynamic layer

43. Which items are examples of localization objects in an HD map?

- A) Traffic congestion and incidents
- B) Traffic signs and poles
- C) Only moving pedestrians
- D) Throttle and brake commands

ANS: B) Traffic signs and poles

44. How do HD maps differ from ordinary navigation maps such as Google Maps?

- A) They contain no road information.
- B) They are intended only for human viewing.
- C) They are updated continuously by every vehicle.
- D) They provide a highly accurate, centimeter-level 3D representation designed for computer processing.

ANS: D) They provide a highly accurate, centimeter-level 3D representation designed for computer processing.

45. How can an HD map reduce the onboard perception workload?

- A) It can reduce the region of interest for detecting traffic elements.
- B) It eliminates all moving objects from the environment.
- C) It increases the camera's field of view.
- D) It replaces vehicle control with cloud computing.

ANS: A) It can reduce the region of interest for detecting traffic elements.

46. Which is a disadvantage of HD maps identified in the presentation?

- A) They cannot include lane geometry.
- B) High acquisition cost and infrequent updates can leave them outdated.
- C) They prevent localization in adverse weather.
- D) They are used only by lower automation levels.

ANS: B) High acquisition cost and infrequent updates can leave them outdated.